Technical SEO Implementation Guide

Next.js / React — Vercel Deployment

Agent-Ready Implementation Reference

Prepared: 26 September 2026 | Framework: Next.js App Router | Host: Vercel

1. Scope & Prerequisites

This guide is written for an AI agent implementing Technical SEO on a Next.js (App Router) website hosted on Vercel. Follow each section in order. Every section ends with an acceptance criterion — verify before moving on.

Stack

- Framework: Next.js 14+ with App Router (app/ directory)
- Deployment: Vercel with automatic CI/CD
- Rendering mix: SSG, ISR, SSR, and Client Components

> *Each section is self-contained. Do not skip sections. Tick the checklist in Section 13 as you go.*

---

2. Metadata API

Next.js App Router uses a typed Metadata API. Every page must export either a static metadata object or a dynamic generateMetadata() function. No manual <head> or <meta> tags are needed.

2.1 Root Layout — app/layout.tsx

Set site-wide defaults here. This is the fallback for all pages that do not override metadata. Required fields: metadataBase, title (with template), description, openGraph, twitter, and robots.

> *metadataBase must be set to your production domain, e.g. new URL("https://yourdomain.com"). Without it, relative OG image paths will not resolve correctly.*

2.2 Static Pages

Each page exports a metadata object with title, description, and alternates.canonical. Title renders through the root template automatically (e.g. "About Us | Site Name").

2.3 Dynamic Pages (Blog / Product)

Use generateMetadata({ params }) to fetch the record and return metadata. Include canonical, OG image, og:type: "article", publishedTime, and modifiedTime for blog posts.

2.4 Rules

- Every page must have a unique title and unique meta description.
- Canonical URL is mandatory on every page via alternates.canonical.
- OG image must be 1200 × 630 px. Store in /public/og/.
- Twitter card type: summary_large_image.

> *Acceptance: Paste any URL into https://opengraph.xyz — title, description, and OG image must all render correctly.*

---

3. XML Sitemap

Create app/sitemap.ts. Vercel auto-serves it at /sitemap.xml. For sites with database-driven content (blog, products), the function must be async and fetch all published records.

3.1 What to Include

- All public, indexable pages with their canonical URLs.
- lastModified date for every URL (use updatedAt from database records).
- changeFrequency and priority for each route type.

3.2 What to Exclude

- API routes (/api/*), admin pages, auth pages, dashboard routes.
- Pages with noindex in their metadata.
- Duplicate URLs (pagination, filter variants).

> *Acceptance: /sitemap.xml returns valid XML. Submit the sitemap URL in Google Search Console → Sitemaps.*

---

4. robots.txt

Create app/robots.ts. Vercel serves it at /robots.txt. The file must reference the sitemap URL and explicitly disallow crawling of internal, API, and auth routes.

Routes to Disallow

- /api/ — all API endpoints
- /admin/ — admin panel
- /dashboard/ — authenticated area
- /_next/ — Next.js internal assets

Optional

Add a separate rule for AI crawlers (GPTBot, ClaudeBot, CCBot) if you wish to block them from training data collection.

> *Acceptance: /robots.txt is publicly accessible. Sitemap URL is present. Disallowed routes return 403 or are inaccessible to crawlers.*

---

5. Canonical URLs & Redirects

5.1 Trailing Slash

Set trailingSlash in next.config.js and enforce it consistently across all pages. Mixing /about and /about/ creates duplicate content. Pick one and never deviate.

5.2 301 Redirects

Define all permanent redirects in next.config.js inside the redirects() async function. Use permanent: true for 301 (passes link equity). Use permanent: false for 302 (temporary, does not pass equity).

- Old domain or old URLs → new URLs: always 301.
- Temporary campaigns or A/B tests: 302.

5.3 WWW vs Non-WWW

Set the canonical domain in Vercel Dashboard → Project → Domains. Vercel handles the 301 redirect automatically. No code required.

> *Acceptance: curl -I https://yourdomain.com/old-page returns HTTP 301 with Location header pointing to the new URL.*

---

6. Performance & Core Web Vitals

Core Web Vitals (LCP, CLS, INP) are confirmed Google ranking signals. The targets are: LCP < 2.5s, CLS < 0.1, INP < 200ms.

6.1 Images — next/image (Mandatory)

Replace every <img> tag with the Next.js <Image> component. Add priority prop to all above-the-fold images (hero, banner). Set appropriate sizes prop for responsive images. Use WebP/AVIF formats by default via next.config.js image formats config.

6.2 Fonts — next/font

Load all fonts through next/font/google or next/font/local. Never use a raw <link> to Google Fonts. The font must be declared in the root layout with display: "swap" to prevent invisible text during load.

6.3 Third-Party Scripts — next/script

All third-party scripts (analytics, tag managers, chat widgets) must use the Script component. Strategy options:

- afterInteractive — analytics, tag managers (loads after page hydration).
- lazyOnload — chat widgets, surveys (loads during browser idle time).
- beforeInteractive — only for scripts that must run before hydration.

6.4 next.config.js Flags

Enable these in next.config.js: compress: true (gzip), poweredByHeader: false (removes X-Powered-By), images.formats: ["image/avif", "image/webp"], experimental.optimizeCss: true (inlines critical CSS).

> *Acceptance: PageSpeed Insights score ≥ 90 on mobile and desktop for all key pages. Run at https://pagespeed.web.dev*

---

7. Structured Data (JSON-LD)

Structured data enables rich results in Google Search (star ratings, breadcrumbs, FAQs). Inject via a <script type="application/ld+json"> tag using dangerouslySetInnerHTML inside the page JSX.

Required Schemas by Page Type

Schema Type	Page(s)	Priority
Organization	Root layout (all pages)	Required
WebSite + SearchAction	Homepage	Required
BreadcrumbList	All inner pages	Required
Article	Blog posts	Required if blog exists
FAQPage	FAQ / support pages	Recommended
Product	Product / service pages	Required if e-commerce
LocalBusiness	Contact / about page	If physical location

Key Rules

- Organization schema goes in the root layout so it appears on every page.
- Never duplicate schema types on the same page.
- Use absolute URLs (https://yourdomain.com/...) in all schema properties.
- datePublished and dateModified must be ISO 8601 format.

> *Acceptance: Test every schema at https://search.google.com/test/rich-results — zero errors required.*

---

8. HTTP Headers (Vercel)

Security headers improve crawl trust and protect against common attacks. Configure via next.config.js headers() function or vercel.json. Both are applied at the edge.

Required Headers

- X-Content-Type-Options: nosniff — prevents MIME sniffing.
- X-Frame-Options: SAMEORIGIN — prevents clickjacking.

- Referrer-Policy: origin-when-cross-origin — controls referrer data.
- Permissions-Policy: restrict camera, microphone, geolocation.
- Content-Security-Policy: whitelist scripts, styles, and image sources.
- X-Robots-Tag: index, follow — on all public pages.

Cache-Control for Static Assets

In vercel.json, set Cache-Control: public, max-age=31536000, immutable for all image, font, CSS, and JS files. This dramatically improves repeat-visit performance.

> *Acceptance: https://securityheaders.com returns grade A or A+. Static assets show Cache-Control: max-age=31536000 in browser DevTools → Network tab.*

---

9. Rendering Strategy

All publicly indexable content must be server-rendered or statically generated. Client-side rendering (CSR) with useEffect data fetching is invisible to crawlers and must not be used for SEO-critical content.

Strategy	Next.js Method	Use For	SEO
SSG	generateStaticParams()	Marketing, blog, docs	✓ Excellent
ISR	export const revalidate = N	Products, listings	✓ Excellent
SSR	force-dynamic / no-cache	Personalised pages	✓ Good
CSR	useEffect + fetch	Dashboards, auth only	✗ Avoid

Rules

- All landing pages, blog posts, service pages → SSG.
- Listing pages with frequent updates → ISR with appropriate revalidate seconds.
- Pages that should not be indexed (dashboard, cart, checkout) → add robots: { index: false } to metadata.

> *Acceptance: curl -A "Googlebot" https://yourdomain.com/your-page returns full HTML content — not an empty div or loading spinner.*

---

10. HTML Semantics & On-Page Rules

10.1 Heading Hierarchy

- One <h1> per page — must contain the primary keyword for that page.
- <h2> for major sections. <h3> for subsections. Never skip levels.

- Headings must describe content — not be used for styling purposes.

10.2 HTML lang Attribute

The <html> tag in app/layout.tsx must include lang="en" (or the appropriate locale). This is required for accessibility compliance and helps search engines understand the language of the content.

10.3 Semantic HTML

Use semantic elements throughout: <header>, <nav>, <main>, <article>, <section>, <aside>, <footer>. Avoid wrapping all content in generic containers without semantic meaning.

10.4 Internal Linking

- Use next/link for all internal navigation — never a plain <a> tag.
- Anchor text must be descriptive — "Web Development Services", not "click here".
- Every important page must be reachable within 3 clicks from the homepage.
- Add a breadcrumb component to all inner pages.

10.5 Image Alt Text

Every <Image> component must have a descriptive alt attribute. Decorative images use alt="" (empty string). Alt text should describe the image content naturally — do not keyword-stuff.

> *Acceptance: Lighthouse Accessibility score ≥ 90. Confirm one H1 per page using browser DevTools. Zero broken internal links.*

---

11. Analytics & Monitoring

11.1 Vercel Analytics & Speed Insights

Install @vercel/analytics and @vercel/speed-insights. Add the <Analytics /> and <SpeedInsights /> components to the root layout inside <body>. Enable in Vercel Dashboard → Project → Analytics.

11.2 Google Search Console

- Verify ownership via the HTML tag method (add to root layout metadata).
- Submit sitemap URL at /sitemap.xml.
- Monitor: Coverage report, Core Web Vitals, Rich Results, Manual Actions.
- Set preferred domain (www vs non-www) in Settings.

11.3 Google Analytics

Install via next/script with strategy="afterInteractive". Do not use a third-party React wrapper — the native Script component is sufficient and more performant.

> *Acceptance: Google Search Console shows "Sitemap successfully fetched." Vercel Dashboard shows real-user Web Vitals data within 24 hours of deployment.*

---

12. Internationalization (If Applicable)

Skip this section if the site is single-language. For multi-language sites, configure i18n in next.config.js with locales and defaultLocale. Set hreflang alternates in generateMetadata using alternates.languages for every language variant of each page.

- Use absolute URLs in hreflang values.
- Include x-default hreflang pointing to the default locale URL.
- Each locale must have its own sitemap entries.

---

13. Agent Implementation Checklist

Tick each item as you complete it. Do not mark done until the acceptance criterion for that section passes.

Metadata

- ☐ Root layout has metadataBase, title template, OG image, Twitter card, and robots fields.
- ☐ Every page has a unique title and unique meta description.
- ☐ Every page has alternates.canonical set to its absolute URL.
- ☐ Dynamic pages use generateMetadata() function.
- ☐ OG image (1200×630) exists for all key pages.

Crawlability

- ☐ app/robots.ts created — /robots.txt is accessible and correct.
- ☐ app/sitemap.ts created — /sitemap.xml lists all public pages.
- ☐ Sitemap submitted in Google Search Console.
- ☐ Admin, API, and auth routes are disallowed in robots.txt.

Performance

- ☐ All <img> tags replaced with next/image.
- ☐ All fonts loaded via next/font with display: swap.
- ☐ All third-party scripts use next/script with correct strategy.
- ☐ PageSpeed Insights ≥ 90 on mobile and desktop.
- ☐ LCP < 2.5s, CLS < 0.1, INP < 200ms.

Structured Data

- ☐ Organization schema in root layout (appears on all pages).
- ☐ Article / Product / FAQ schema on relevant pages.
- ☐ BreadcrumbList schema on all inner pages.
- ☐ All schemas pass https://search.google.com/test/rich-results with zero errors.

Technical Config

- ☐ Trailing slash enforced consistently in next.config.js.
- ☐ All old/changed URLs have 301 redirects in next.config.js.
- ☐ Security headers pass https://securityheaders.com with grade A or A+.
- ☐ HTTPS enforced — no mixed content on any page.
- ☐ All public content pages use SSG or ISR (not CSR).
- ☐ HTML lang attribute set on root <html> element.

Monitoring

- ☐ @vercel/analytics and @vercel/speed-insights installed and rendering.
- ☐ Google Search Console verified and sitemap submitted.
- ☐ Google Analytics or Tag Manager installed via next/script afterInteractive.

END OF DOCUMENT

Technical SEO Implementation Guide — Next.js / Vercel